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# Abstract

BricsCoin (BRICS) is a decentralized cryptocurrency built on the proven SHA256 Proof-of-Work consensus mechanism, enhanced with post-quantum cryptographic security. Designed for hardware mining compatibility, BricsCoin enables anyone with ASIC mining equipment to participate in securing the network and earning rewards. With a fixed supply of 21 million coins, ultra-low transaction fees of 0.000005 BRICS (burned), and a hybrid ECDSA + ML-DSA-65 signature scheme providing quantum resistance, BricsCoin aims to be a fair, transparent, secure, and future-proof digital currency.

BricsCoin goes beyond a simple payment network by offering a suite of on-chain applications: **BricsChat** (quantum-proof encrypted messaging), **Time Capsule** (time-locked on-chain data), **BricsNFT** (PQC-signed certificates), and an **AI Oracle** (GPT-5.2 powered network intelligence).


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# 1. Introduction

## 1.1 Background

Since Bitcoin's inception in 2009, Proof-of-Work has proven to be the most secure and decentralized consensus mechanism for digital currencies. BricsCoin builds upon this foundation, implementing a clean SHA256-based blockchain optimized for modern ASIC mining hardware.

## 1.2 The Quantum Threat

The development of large-scale quantum computers poses a significant threat to current cryptographic systems. Shor's algorithm can break ECDSA and RSA, the foundations of most blockchain security. BricsCoin proactively addresses this threat by implementing a hybrid post-quantum cryptographic scheme.

## 1.3 Vision

BricsCoin's mission is to create a truly decentralized currency that:

- Remains accessible to hardware miners worldwide
- Maintains ultra-low transaction fees for all users (0.000005 BRICS, burned)
- Provides transparent, verifiable transactions
- Offers quantum-resistant security through ML-DSA-65
- Hosts on-chain applications (Chat, Certificates, Time Capsules)
- Operates as a fully open-source project

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## 2. Technical Specifications

| Parameter | Value |
|---|---|
| Algorithm | SHA256 Proof-of-Work |
| Consensus | Proof-of-Work |
| Max Supply | 21,000,000 BRICS |
| Block Reward | 50 BRICS (initial) |
| Halving Interval | Every 210,000 blocks |
| Target Block Time | ~10 minutes |
| Difficulty Adjustment | Every 2016 blocks |
| Transaction Fee | 0.000005 BRICS (burned) |
| Fee Model | Deflationary (all fees destroyed) |
| Signature Algorithm | ECDSA secp256k1 + ML-DSA-65 (hybrid) |
| Address Format | Legacy: `BRICS` + 40 hex / PQC: `BRICSPQ` + 38 hex |
| Mining Protocol | Stratum v1 |
| License | MIT |

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## 3. Blockchain Architecture

### 3.1 Block Structure

Each block contains:

- **Index**: Sequential block number
- **Timestamp**: UTC block creation time
- **Transactions**: List of validated transactions
- **Previous Hash**: SHA256 hash of the previous block
- **Nonce**: Proof-of-Work solution
- **Difficulty**: Current mining difficulty target
- **Miner Address**: Address that receives the block reward
- **PQC Signature**: Hybrid ECDSA + ML-DSA-65 block signature (v2.0+)

### 3.2 Transaction Structure

Each transaction includes:

- **Sender/Recipient**: BRICS or BRICSPQ addresses
- **Amount**: Transfer amount (max 8 decimal places)

- **Timestamp**: Transaction creation time
- **Signature**: ECDSA digital signature (signed client-side)
- **Public Key**: Sender's ECDSA public key for verification

### 3.3 Mining

BricsCoin uses the same SHA256 algorithm as Bitcoin, ensuring compatibility with existing ASIC mining hardware. Miners connect via the Stratum protocol on port 3333.

**Difficulty adjustment** occurs every 2016 blocks, targeting a 10-minute block time. The algorithm increases difficulty when blocks are mined too quickly and decreases it when blocks are too slow.

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## 4. Post-Quantum Cryptography

### 4.1 The Hybrid Approach

BricsCoin implements a **hybrid signature scheme** combining:

- **ECDSA (secp256k1)**: The classical algorithm used by Bitcoin
- **ML-DSA-65 (FIPS 204)**: The NIST-standardized post-quantum signature algorithm (formerly known as Dilithium)

This dual approach ensures:

- **Backward compatibility**: Legacy wallets continue to function
- **Quantum resistance**: ML-DSA-65 remains secure against quantum attacks
- **Defense in depth**: Even if one algorithm is compromised, the other provides security

### 4.2 Client-Side Signing

A key architectural decision in BricsCoin is that **all cryptographic signing occurs in the user's browser**:

1. Private keys are generated locally using `@noble/post-quantum`
2. Transactions are signed client-side before being submitted to the network
3. Private keys are **never** transmitted to any server
4. The server only verifies pre-signed transactions

This eliminates the most common attack vector in cryptocurrency platforms: server-side key compromise.

### 4.3 PQC Wallet Addresses

PQC wallets use the `BRICSPQ` prefix to distinguish them from legacy wallets:

- Legacy: `BRICS` + SHA256(public_key)[:40] = 45 characters
- PQC: `BRICSPQ` + SHA256(public_key)[:38] = 45 characters

### 4.4 Block Signing

Starting with v2.0, each node maintains its own PQC key pair. When a node mines a new block, it signs the block with both its ECDSA and ML-DSA-65 private keys. This provides verifiable attribution and integrity for all blocks on the chain.

### 4.5 Migration

Users can migrate from legacy ECDSA wallets to PQC wallets at **zero cost** (no transaction fee). The migration endpoint `/api/pqc/migrate` transfers the full balance from a legacy address to a new PQC address.

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## 5. Tokenomics

### 5.1 Supply

- **Max Supply**: 21,000,000 BRICS (hard cap)
- **Block Reward**: 50 BRICS (halves every 210,000 blocks)
- **Premine**: Genesis block allocation for development

### 5.2 Fee Model (Deflationary)

Transaction fees of **0.000005 BRICS** are **burned** (permanently destroyed), creating a deflationary pressure on the supply over time. This mechanism:

- Reduces circulating supply with each transaction
- Provides anti-spam protection
- Aligns incentives for long-term holders

All on-chain features (BricsChat, Time Capsule, BricsNFT) use the same burn fee, contributing to the deflationary model.

### 5.3 Halving Schedule

| Event | Block | Reward |
|---|---|---|
| Genesis | 0 | 50 BRICS |
| 1st Halving | 210,000 | 25 BRICS |

| | | |
|---|---|---|
| 2nd Halving | 420,000 | 12.5 BRICS |
| 3rd Halving | 630,000 | 6.25 BRICS |
| ... | ... | ... |

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# 6. On-Chain Applications

BricsCoin is more than a payment network. It hosts a suite of on-chain applications that leverage PQC signatures and the burn-fee mechanism.

## 6.1 BricsChat — Quantum-Proof On-Chain Messaging

BricsChat is the **world's first PQC-encrypted on-chain messaging system**. Each message is:

- Signed with the sender's hybrid ECDSA + ML-DSA-65 keys
- Stored immutably on the blockchain
- Publicly visible in the Global Feed (read-only for visitors)
- Accompanied by a **0.000005 BRICS** fee that is burned

**How it works:**

1. User connects their PQC wallet
2. Writes a message and specifies a recipient address
3. The message content is hex-encoded and signed client-side
4. A burn transaction is created on-chain
5. The message is permanently recorded with the sender's PQC signature

**Use cases:** Immutable declarations, public statements, provable communication timestamps.

## 6.2 Decentralized Time Capsule

The Time Capsule feature allows users to store encrypted data on-chain that becomes accessible only at a specific future block height.

- Data is locked until the target block is mined
- Each capsule creation burns **0.000005 BRICS**
- Capsules can be public or addressed to a specific recipient
- Content is immutable once locked

**Use cases:** Timed announcements, future predictions, proof-of-knowledge at a specific time, community events.

## 6.3 BricsNFT — PQC-Signed On-Chain Certificates

BricsNFT is the **world's first NFT system with post-quantum cryptographic signatures**. It allows anyone to mint immutable certificates on the BricsCoin blockchain.

**Certificate Types:**

- Diploma / Degree
- Property Deed
- Authenticity Certificate
- Professional License
- Membership
- Award / Achievement
- Software License
- Custom (user-defined)

**How it works:**

1. Issuer connects their PQC wallet
2. Selects a certificate type (or creates a custom one)
3. Fills in title, description, and optionally a recipient address
4. The certificate content is SHA-256 hashed and signed with ECDSA + ML-DSA-65
5. A burn transaction of **0.000005 BRICS** is created
6. The certificate receives a unique ID (e.g., `BRICSNFT-A1B2C3D4E5F6`)
7. Anyone can verify the certificate's authenticity using the Verify tool

**Trust Model:**

The blockchain records **who** signed **what** and **when** — immutably. To verify the issuer's identity, users check that the issuer's PQC address matches the one published on their official website or channels. This is the same trust model used by digital signatures and PEC (certified email).

**Features:**

- Public gallery of all minted certificates
- Certificate transfer between PQC addresses
- On-chain verification tool
- Transfer history tracking
- Deflationary burn fee

## 6.4 AI Blockchain Oracle

The AI Oracle is powered by **GPT-5.2** and provides real-time network intelligence:

- **Network Analysis**: Health score, mining analysis, security assessment, recommendations
- **Predictions**: Difficulty trend, hashrate forecast, halving impact, network growth outlook
- **Ask Oracle**: Conversational AI that answers questions about BricsCoin using live network data

The Oracle analyzes real blockchain data including block times, difficulty, miner distribution, transaction volume, and PQC adoption rates.

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# 7. Network Architecture

## 7.1 API Server

The BricsCoin API is built on FastAPI (Python) with:

- Asynchronous MongoDB (Motor) for high-performance database operations
- Rate limiting to prevent abuse
- Security headers and CORS configuration
- RESTful endpoints for all blockchain operations
- Dedicated routers for Chat, Time Capsule, NFT, and Oracle features

## 7.2 Mining Server

The Stratum server supports:

- Stratum v1 protocol (port 3333)
- Variable difficulty per worker
- Share validation and hashrate tracking
- PQC block signing on successful mine

## 7.3 Peer-to-Peer Network

Nodes communicate via HTTP API calls for:

- Block propagation
- Transaction broadcasting
- Blockchain synchronization
- Peer discovery

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# 8. Security

## 8.1 Live Security Audit

BricsCoin includes a built-in security audit system that runs 27 real-time tests covering:

- Input validation (8 tests)

- Classical cryptography (5 tests)
- Post-quantum cryptography (6 tests)
- Attack prevention (8 tests)

The audit can be executed at any time via `GET /api/security/audit`.

### 8.2 Attack Mitigations

| Attack | Mitigation |
|---|---|
| Replay Attack | Signature uniqueness + timestamp validation |
| 51% Attack | SHA256 PoW (same security model as Bitcoin) |
| Sybil Attack | Proof-of-Work requirement |
| DDoS | Rate limiting + IP blacklisting |
| Quantum Attack | ML-DSA-65 hybrid signatures |
| Key Theft | Client-side signing (keys never leave device) |

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## 9. Roadmap

### Completed

- SHA256 PoW blockchain
- ECDSA wallets and transactions
- Stratum mining protocol
- Web interface (Dashboard, Explorer, Wallet)
- Post-Quantum Cryptography (ML-DSA-65)
- Hybrid ECDSA + PQC signatures
- Client-side browser signing
- Zero-fee wallet migration
- Live security audit (27/27 tests)
- BricsChat — On-chain PQC-encrypted messaging
- Decentralized Time Capsule
- BricsNFT — PQC-signed on-chain certificates
- AI Blockchain Oracle (GPT-5.2)
- Deflationary burn-fee mechanism
- Production deployment

### Future

- Mobile wallet application
- Stratum v2 protocol
- P2Pool decentralized mining
- Lightning-style payment channels

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## 10. Conclusion

BricsCoin combines the proven security of SHA256 Proof-of-Work with forward-looking post-quantum cryptographic protection. By implementing ML-DSA-65 alongside ECDSA in a hybrid scheme, BricsCoin is positioned to remain secure even as quantum computing technology advances.

Beyond a simple payment network, BricsCoin offers a complete ecosystem of on-chain applications — from quantum-proof messaging to PQC-signed certificates — all contributing to a deflationary token economy through burn fees.

The commitment to client-side signing, open-source development, and community-driven governance ensures that BricsCoin remains transparent, accessible, and trustworthy.

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3. Bernstein, D.J. et al. "CRYSTALS-Dilithium: A Lattice-Based Digital Signature Scheme"
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**Website**: bricscoin26.org

**Repository**: codeberg.org/Bricscoin_26/Bricscoin

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